

SPATIAL ANALYSIS REPORT

Gram Panchayat Distribution — Raigad District

Maharashtra, India

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Based on Bhuvan (ISRO) Administrative Boundaries

State: Maharashtra (Code 27)

District: Raigad (Code 520)

Extent: 17.88°N–19.11°N, 72.87°E–73.64°E

Date: May 2026

KEY FIGURES			
761	14	6,053	7.95
Gram Panchayats	Blocks (Talukas)	Area (km ²)	Avg GP Area (km ²)

1. Overview

Raigad district lies along the Konkan coast of Maharashtra, stretching approximately **81 km east–west** and **136 km north–south**. It encompasses **761 Gram Panchayats (GPs)** across **14 blocks**, with a combined polygon area of approximately **6,053 km²**. The district is bounded by the Arabian Sea to the west, the Western Ghats escarpment to the east, and borders Pune and Thane districts inland.

Spatial boundaries were derived from the Bhuvan portal maintained by ISRO’s National Remote Sensing Centre (NRSC), representing the most comprehensive publicly available village-level administrative boundary dataset for India.

Extent: 17.88°N to 19.11°N latitude, 72.87°E to 73.64°E longitude. The district’s elongated north–south geometry (aspect ratio $\approx 1 : 1.7$) reflects the narrow Konkan coastal strip compressed between the sea and the Sahyadri ranges.

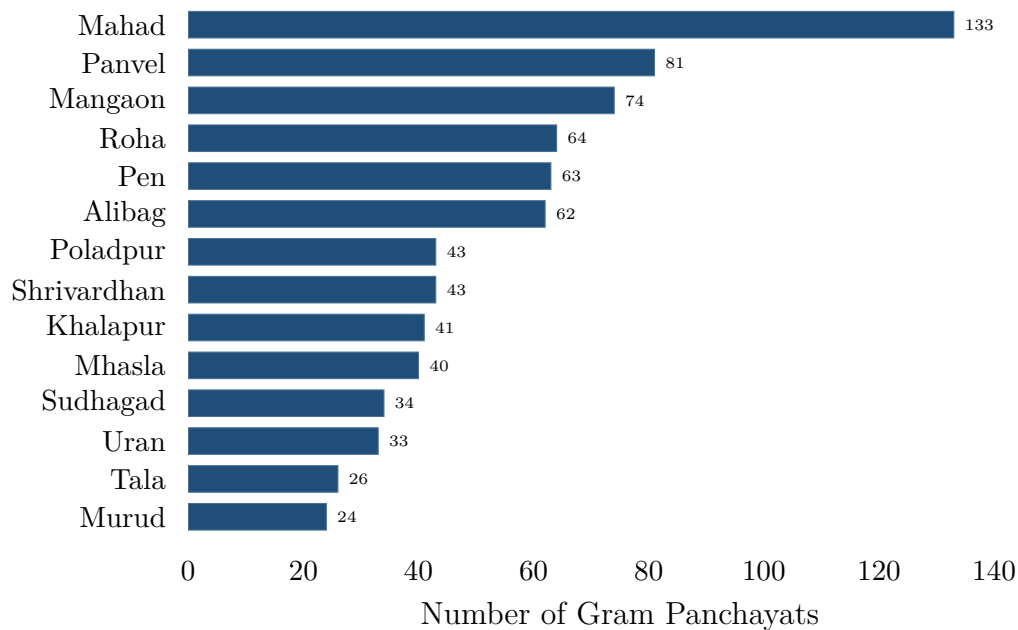
2. Block-level Summary

Mahad is the largest block by both GP count (133 GPs, 17.5% of the total) and area ($\approx 814\text{ km}^2$), reflecting its interior position against the Western Ghats. Uran, despite having only 33 GPs, represents one of the most economically significant blocks due to its proximity to Navi Mumbai and the JNPT port. Sudhagad has the largest average GP area (13.62 km^2), indicating sparser administrative subdivision consistent with its hilly, forested terrain.

Block	Code	GPs	Area (km ²)	Avg GP (km ²)
Mahad	4185	133	814.0	6.12
Panvel	4173	81	522.3	6.45
Mangaon	4181	74	684.4	9.25

Block	Code	GPs	Area (km ²)	Avg GP (km ²)
Roha	4179	64	631.2	9.86
Pen	4176	63	491.5	7.80
Alibag	4177	62	483.6	7.80
Poladpur	4186	43	364.8	8.48
Shrivardhan	4183	43	247.0	5.74
Khalapur	4175	41	372.0	9.07
Mhasla	4184	40	309.1	7.73
Sudhagad	4180	34	463.0	13.62
Uran	4172	33	185.6	5.62
Tala	4182	26	239.3	9.20
Murud	4178	24	245.5	10.23
Total	—	761	6,053.3	7.95

3. GP Count by Block



4. Spatial Patterns & Observations

Size–density inversion. There is a clear inverse relationship between average GP area and GP density. Coastal and peri-urban blocks (Uran: 5.62 km²/GP; Shrivardhan: 5.74 km²/GP) show smaller, more densely packed GPs reflecting higher population pressure, while interior blocks like Sudhagad (13.62 km²/GP) and Murud (10.23 km²/GP) have larger, sparser units consistent with forested or topographically complex terrain.

North–south gradient. Northern blocks — Uran, Panvel, Khalapur — lie within the Mumbai Metropolitan Region influence zone and show compact GP geometries indicative of intensified land use. Southern blocks — Mahad, Poladpur, Mangaon — extend into the Sahyadri foothills and contain larger, more irregular polygons corresponding to dense forest cover and river valley settlements.

Coastal strip. Alibag, Murud, and Shrivardhan blocks form a continuous coastal strip along the Arabian Sea. Despite moderate GP counts, these blocks are spatially narrow and elongated, with GPs following the shoreline and creek systems of Raigad’s deeply indented Konkan coast.

Block coverage. The combined GP polygon area of $\approx 6,053 \text{ km}^2$ compares closely to the official district area of $7,152 \text{ km}^2$ (Census 2011). The $\approx 15\%$ gap is expected and attributable to urban local body areas (Panvel Municipal Corporation, Alibag Municipal Council, Uran Nagar Panchayats) excluded from the GP-level dataset, plus minor gaps at inter-state boundaries.

5. Delivered Data Outputs

File	Format	CRS	Recs	Use
raigad_panchayats.geojson	GeoJSON	WGS84 (EPSG:4326)	761	GIS / web mapping
raigad_panchayats.xlsx	Excel (.xlsx)	—	761	Tabular analysis
raigad_panchayats.csv	CSV (UTF-8)	—	761	Data pipelines

The GeoJSON layer uses **WGS84 (EPSG:4326)** geographic coordinates. Geometry type is MultiPolygon. Attribute fields: `s_name`, `s_code`, `d_name`, `d_code`, `b_name`, `b_code`, `gp_name`, `gp_code`.

6. Data Source & Citation

Primary Source:

Bhuvan Geoportal, National Remote Sensing Centre (NRSC), Indian Space Research Organisation (ISRO). *India Administrative Boundaries — Gram Panchayat Level*. Distributed via <https://github.com/yashveeeeeer/india-geodata/tree/main/data/administrative/panchayats> as `bhuvan_panchayats.geojsonl.7z`. Format: GeoJSONL (7z compressed). Accessed May 2026.

Suggested citation:

ISRO/NRSC (May, 2026). *Bhuvan Panchayat Boundaries* [GeoJSONL dataset]. Retrieved from <https://github.com/yashveeeeeer/india-geodata>, accessed May 2026. Subset: Raigad District, Maharashtra (district code 520), 761 GPs across 14 blocks.

Raigad GP Spatial Report

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Data: Bhuvan / ISRO-NRSC | Generated May 2026